MAYANK JEET

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Professional Summary

B.Tech Civil Engineering student at IIT Bhubaneswar with experience spanning full-stack software engineering, applied machine learning, agentic AI systems, and data analysis. Solved 600+ DSA problems in C++ (LeetCode | CodeForces) and built and deployed production-grade AI applications — an autonomous task-executing agent, a RAG-based content pipeline, and a computer-vision defect-detection system (LangGraph, FastAPI, React, PyTorch).

Key Projects

BlogGEN: RAG-Based Blog Generation App | LangGraph, FastAPI, React, Tavily **- GitHub**

- Built and deployed a full-stack app that turns a topic into a researched, structured blog: a LangGraph pipeline breaks the topic into tasks, retrieves sources per task via on-demand web research (Tavily), and writes the final post.
- Designed a token-budget monitor that allocates research depth per task by importance, keeping LLM usage under control, plus automatic image placement: the model decides where visuals are needed and generates them (Gemini).
- Built a ChatGPT-style React interface with a sidebar to switch between past chats and start new ones, backed by per-thread state in a LangGraph checkpointer. FastAPI backend on Render, frontend on Vercel.

OpsPilot: Autonomous AI Task Worker with Human Approval | LangGraph, FastAPI, Playwright, React **- GitHub**

- Built and deployed an AI agent that turns a plain-English task into a verified business workflow: a LangGraph plan-act-observe loop where the LLM picks one structured action at a time from 8 tools, while deterministic guards enforce step order and rules.
- Automated a simulated accounting app through a real headless browser (Playwright); the agent independently verifies the ledger result and detects duplicates to avoid double-posting.
- Designed a human-in-the-loop safety gate: high-value invoices pause for approve/reject, enforced in both the executor and the API, so the model cannot approve its own transaction. Added retry-based failure recovery and multi-currency parsing with live exchange rates.

Anomaly-Inspect | PyTorch, FAISS, ONNX Runtime, FastAPI, React **- GitHub**

- Built an unsupervised defect-detection system on MVTec AD, comparing a convolutional-autoencoder baseline against a PatchCore-style method (frozen ResNet-18 features + FAISS memory banks) across 15 categories.
- Reached **0.977 mean image AUROC** / **0.973 mean pixel AUROC** with the PatchCore-style method, versus 0.669 / 0.505 for the autoencoder baseline.
- Exported the model to ONNX and deployed it as a FastAPI + React demo with ~97ms median end-to-end latency on CPU.

Experience

Core Head - Sports Fest, IIT Bhubaneswar

- Lead a team of organizers for the fest, assigning task ownership and keeping event operations on schedule.
- Handle cross-functional planning and logistics for large-scale campus event execution.

Technical Skills

Programming	Python, C++, C, SQL (MySQL), HTML, CSS, JavaScript
Machine Learning / AI	PyTorch, TensorFlow, Keras, ONNX Runtime, FAISS
Generative AI	LangChain, LangGraph, RAG, LLM APIs (OpenAI, Anthropic, Gemini)
Data Science	Pandas, NumPy, Matplotlib, Seaborn, Statistical Analysis
Web Development	FastAPI, Streamlit, React, REST APIs, Model Deployment
Tools & Platforms	Git, GitHub, Jupyter Notebook, VS Code, Google Colab, Render, Vercel, Docker
Specializations	Machine Learning, Deep Learning, Computer Vision, Data Analytics, Data Science

Practice Repositories: LeetCode (C++) · LeetCode (Python) · SQL Practice

Competitive Programming: LeetCode · CodeForces

Education

Qualification	Institute	Board/ University	Passing Yr.	Marks	Remarks
B.Tech (Civil Engg.)	IIT Bhubaneswar	Board of Governors (BoG)	2028	8.46 CGPA	Pursuing
Class XII	PBPS, Gaya	CBSE	2024	88.4%	Passed
Class X	DAV PS, Gaya	CBSE	2022	89.4%	Passed